



Transparent superhydrophilic cellulose-based coating with anti-fogging, anti-biofouling and oil/water separation properties

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ABSTRACT

It is a huge challenge to design multifunctional superhydrophilic coatings by bio-based raw materials and low carbon emissions in full life cycle. As a green, hydrophilic, low-cost material, cellulose has irreplaceable advantages in the preparation of sustainable superhydrophilic coatings. Here, a novel and large-scale transparent saxifrage-biomimetic superhydrophilic cellulose-based coating (SES) was fabricated by one-step method through blending semi-wet maleic anhydride esterified cellulose (EC) modified cellulose with waterborne styrene-acrylate latex (SAL) and sodium dodecyl sulfate (SDS). The EC was chosen as fillers because of its smallest particle size and the best dispersion of cellulose in waterborne latex. With the addition of SDS, affected by polarity, the hydrophilic head group orientally arranges the cellulose to form a saxifrage-like structure on the surface of coating. The obtained superhydrophilic coating with a static water contact angle of 0° and a waterborne oil contact angle of 157.1°, and the coating remained in a superhydrophilic state after 600 times of wear resistance cycle test. The SES coatings have excellent performance in anti-fogging, biological anti-fouling and oil-water separation, this is due to the synergy effects of the micro-nano scale roughness, high surface energy and hydrophilic group of coating film. The life cycle assessment (LCA) analysis of saxifrage-biomimetic SES superhydrophilic coating proved that the cellulose-based coatings were low-cost, easily prepared, and non-pollution, reducing carbon emission.

1. Introduction

As the current global focus shifts towards environmental protection and carbon reduction, development of coating materials from bio-derived sources has attracted extensive attention [1] [2]. In the past few years, production routes from renewable sources have become more robust. Cellulose extracted from wood, cotton, ramie, straw and bagasse, reigns as the most abundant organic compound on Earth, which is renewable, degradable and has high mechanical strength [3–4]. Cellulose is an unbranched linear polymer with many polar hydroxyl groups on its molecular chains, which can enhance the mechanical properties [5] and surface wettability [6] of composite coatings. In the light of this, numerous researches have demonstrated the potential to substitute cellulose for organic/inorganic fillers, especially petrochemical-based one, in the functional coating materials [2], [4], [7]. However, it is a huge challenge to reduce the high hygroscopicity of cellulose due to

excess hydroxyl groups, leading to coating swelling, which decreasing the binding force between cellulose and coating [6]. As well as micro-nano particle. Ultrasound dispersion after esterification reaction, is a clean and energy-saving pretreatment method to consumes hydroxyl groups and reduces the size of cellulose and consume hydroxyl group [8]. Excessed hydroxyl groups on cellulose can be esterified with acids, thereby introducing cellulose esters with specific properties on the surface of cellulose, breaking the hydrogen bonds between cellulose molecules and promoting their dispersion. In addition, alkali treatment can effectively improve the roughness of the fiber surface, which is a method to improve the interfacial strength between the cellulose and the coating. These fiber pretreatments are particularly important for the preparation of hydrophilic coatings.

Superhydrophilic coatings, which have a water contact angle of less than 5°, have high surface energies, because they create favorable interactions with water [9]. This superior performance effectively

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addresses contaminations on the coating surface, preventing economic losses caused by coating failure. Therefore, they have a wide range of practical applications in self-cleaning [10], anti-fogging [11], sewage treatment [12–13]. What's more, with the rising demand for global energy, accidents such as oil spills and industrial discharges of oily wastewater frequently occur [14]. Recently, extreme surface wettability has achieved great success in oil/water separation and anti-fouling [15]. These examples demonstrate the vast potential and broad application prospects of superhydrophilic coatings. To achieve superhydrophilic coating, many preparation methods have been adopted for hydrophilic groups and certain surface structure [16–23]. For instance, surface roughness has been improved through the addition SiO_2 [17,18] and TiO_2 [19], which are generally regarded as inorganic fillers. On the other hand, surfactant was utilized to higher surface energy such as sodium olefin sulfonate [20]. These studies focus on solvent organic coatings or undegradable raw materials to prepare superhydrophilic coatings. Bio-based superhydrophilic coatings, like chitosan coatings, facing problem of low mechanical strength [22]. The waterborne cellulose coatings is the development direction of environmentally friendly coatings [1]. However, it is less reported about cellulose superhydrophilic coatings because the high moisture absorption of cellulose will cause swelling and poor water-resistance in waterborne coatings.

In nature, many plants have evolved hydrophilic surface to capture water droplets. The hydrophilicity of plant surfaces is primarily determined by the structure and morphology of epidermal cells, the morphology of waxes, tomentum, stomata and surface chemical composition [24]. These plants usually have either well-arranged structure and hydrophilic surface, such as *Nepenthes* [25], *Cactus Spines* [26] and *Saxifrage* [27]. Researchers have been inspired by these natural plants and designed superhydrophilic coatings for various applications [28–31]. Among these plants, *Saxifrage* is a kind of plant with hierarchical structures and close-set tomenta covering its surface. The tomentum structure and hydrophilic surface enable *Saxifrages* to effectively capture waterdrops and absorb them rapidly. Taking inspiration from *Saxifrage* with tomentum-structure hydrophilic surface, superhydrophilicity can be achieved through the design of specific surface structures and coatings with high surface energy.

Motivated by the abovementioned strategy, in this study, *Saxifrage*-inspired coatings with a superhydrophilic surface was fabricated. By comparing the modification of cellulose by wet esterification, semi-wet esterification, alkali treatment and alkali-esterification modification, semi-wet esterification was finally selected. The size of cellulose was miniaturized and excess hydroxyl groups on cellulose were converted into ester groups [32] to improve the dispersity and hydrophilicity in waterborne coating system. The semi-wet esterificated cellulose (EC) emulated the tomentum structure of *Saxifrage* to provide certain surface roughness. EC was added to the aqueous styrene acrylic latex (SAL) with sodium dodecyl sulfate (SDS) to obtain SAL/EC/SDS (SES) composite coating. SDS with a hydrophilic and a hydrophobic end acted as the surfactant to promote the SES coating roughness and surface energy, and made the cellulose arranged directionally in coating. The SES coating exhibited excellent superhydrophilicity, underwater superoleophobicity, and high abrasive resistance. Table S1 indicated the comparison of SES and other superhydrophilic coatings. The study provides a novel example of environmentally friendly coatings, which can be used in antifogging, antifouling and oil/water separation.

2. Experimental section

2.1. Materials

Cellulose, purchased from Green Building Technology Co., Ltd. Maleic anhydride (99.5 %) was purchased from Aladdin Industrial Corporation, Sodium hydroxide (NaOH) (AR) and Sodium dodecyl sulfate (AR) was purchased from Sinopharm Chemical Reagent. Styrene acrylic latex (CP) was purchased from Luyuan Chemical. *Cyclotella*

cryptica (CC) (10^5 mL^{-1}) was purchased from Shanghai Guangyu Biotechnology Technology Ltd. Fluorescein isothiocyanate-labelled bovine serum albumin (FITC-BSA) was provided by Solarbio Life Sciences Ltd. (China). Artificial seawater (ASW) was prepared according to ASTM D1141–98. Probe liquids for underwater oleophobicity and water-oil separation: glycerol, kerosene, silicone oil and hexadecane were commercial products. Methyl blue and oil red dyed with water and oil were purchased from Aladdin Industrial Corporation. Deionized water (DI) was used throughout all experiments.

2.2. Cellulose pretreatment

2.2.1. Wet maleic anhydride esterification

In a three-necked flask, 1 g cellulose with 10 mL deionized water and 5 g maleic anhydride was stirred at 120°C for 4 h, 60 rpm. Then excess maleic acid was neutralized by 1 % volume solution of NaOH until the pH reached 11.0. Excess NaOH is then removed by washing with deionized water until the dispersion is essentially neutral. Esterified cellulose was dispersed by a 40 kHz sonicator. Processing parameters: 1 s/2 s on/off pulse with a power of 700 W and a processing time of 10 min.

2.2.2. Semi-wet maleic anhydride esterification

Reduced the amount of deionized water to 5 mL, and the other steps are the same as 2.2.1.

2.2.3. NaOH alkali treatment

In a beaker, 1 g cellulose with 100 mL deionized water and 3 g NaOH was stirred at 25°C for 4 h, 60 rpm. Then excess NaOH was neutralized by 1 % volume solution of citric acid until the pH reached essentially neutral. Cellulose micro-nanoized by a 40 kHz sonicator. Processing parameters: 1 s/2 s on/off pulse with a power of 700 W and a processing time of 10 min.

2.2.4. Alkali-esterification modification

Proceeded to steps 2.2.3 and 2.2.2 in sequence.

2.3. Preparation of waterborne bio-based superhydrophilic coatings

According to the volume ratio of 4 parts of 1 wt% fiber dispersion and 6 parts of styrene-acrylic latex, stir and blend, and then add 6 wt% SDS. Using a 150- μm wire rod to the glass slide and placed in a vacuum oven at 80°C for 24 h. The thicknesses of dry coating films were $25 \pm 2 \mu\text{m}$. The detail process parameters of coating were placed in Supporting Information 1 (Fig. S1).

2.4. Preparation of oil-water separated mesh

The 2000-mesh nylon mesh was cleaned ultrasonically with water and then dried at room temperature. The mesh was dipped directly into the SES hybrid solution. Then the coated mesh was dried in a vacuum oven at 80°C to give superhydrophilic and underwater superoleophobic membranes suitable for oil-water separation.

2.5. Characterization

The particle size of the cellulose and the uniformity of dispersion in aqueous coatings were observed by optical microscope (OM, BX63, Olympus, Japan). Fourier-transform infrared (FT-IR) spectra were obtained on a Nicolet 6700 spectrometer (Thermo-Fisher Scientific, CA, America) equipped with transmission sampling accessories. Morphology was observed by a field-emission scanning electron microscope (FE-SEM, Ultra55, Carl Zeiss, Germany), at an accelerated voltage of 3 kV in secondary electron mode. All the samples were coated with gold by a sputtering coater (Q150R, Quorum, England) prior to observation. Water contact angel (WCA) and underwater oil contact angel (UWOCA)

were measured with a contact angel goniometer (OCA25, Data Physics, Germany). All measurements were carried out by injecting a 2 μL probe droplet with a screw-top syringe, and average values were obtained from three different regions of the same sample. UV–Vis transmittance spectra were tested with resolution of 2 nm on an UV/Vis/NIR spectrophotometer (U-4100, Hitachi, Japan) equipped with an integrating sphere. The mechanical properties of the SES films were tested using a universal testing machine (5966, Instron, America). The abrasive wear resistance of SES coating was tested by using a standard linear Taber Abrasion apparatus with a CS-5 abrader (352, Taber, America). The adhesion test refers to the GB/T-1720-1979 Paint Film Adhesion Determination Method.

2.6. Antifogging test

Three pieces of glass were uncoated and coated with pure SAL, SES coating. For the humidifying condition, the glass was 3 cm above water which was heated to 100 $^{\circ}\text{C}$. Glasses, car mirrors, goggles and solar panels are tested in the same way.

2.7. Anti-biofouling test

2.7.1. Protein adhesion assays

The anti-protein property was examined using FITC-BSA as a model protein foulant. Coating coupons ($38 \times 26 \text{ mm}^2$) were mounted into cell-culture dishes. FITC-BSA solution (4 mL, 20 $\mu\text{g mL}^{-1}$ in phosphate-buffered saline (PBS), pH 7.4) was injected into each dish and incubated at 25 $^{\circ}\text{C}$ for 7 h in dark. Then, the solution was discarded and the panels were rinsed with fresh PBS buffer for five times to remove the free FITC-BSA. Confocal laser scanning microscopy (CLSM) images were

recorded on a C^{2+} confocal microscope (Nikon, Japan), equipped with 488 nm laser excitation.

2.7.2. Marine algae adhesion assays

Using *cyclotella cryptica* (CC) as model marine algae. Coating coupons ($38 \times 26 \text{ mm}^2$) were mounted into cell-culture dishes. CC (105 algae mL^{-1}) was added into each dish and incubated under the lights for 48 h. Then, the panels were rinsed with 50 mL autoclaved filtered seawater (AFSW) to remove the free algae. Confocal laser scanning microscopy (CLSM) images were recorded on a C^{2+} confocal microscope (Nikon, Japan), equipped with 488 nm laser excitation.

3. Results and discussion

3.1. Comparison of cellulose pretreatment

The results of several cellulose-modified fibers are shown in Fig. 1(a). Untreated cellulose fibers were large and entwined, and it is difficult to disperse uniformly (Fig. 1(a) (i)). Because cellulose is a polysaccharide with a large number of hydroxyl groups (OH), interacted with water to form clusters. The cellulose entanglement modified by wet maleic anhydride esterification was significantly less, and it was relatively short and dispersed. While semi-wet maleic anhydride esterified cellulose achieved smaller particles and evenly dispersed in waterborne coating system. Alkali-treated cellulose has a little rough edge, but there is also a lot of cellulose winding. (Fig. 1(a-b)). Since the fiber surface research produces a large number of hydroxyl groups, it is easy to agglomerate and is not easy to disperse. While maleic anhydride can react with three hydroxyl groups carried by each glucose residue ring in the cellulose molecule, and formed different types of cellulose esters. Cellulose esters

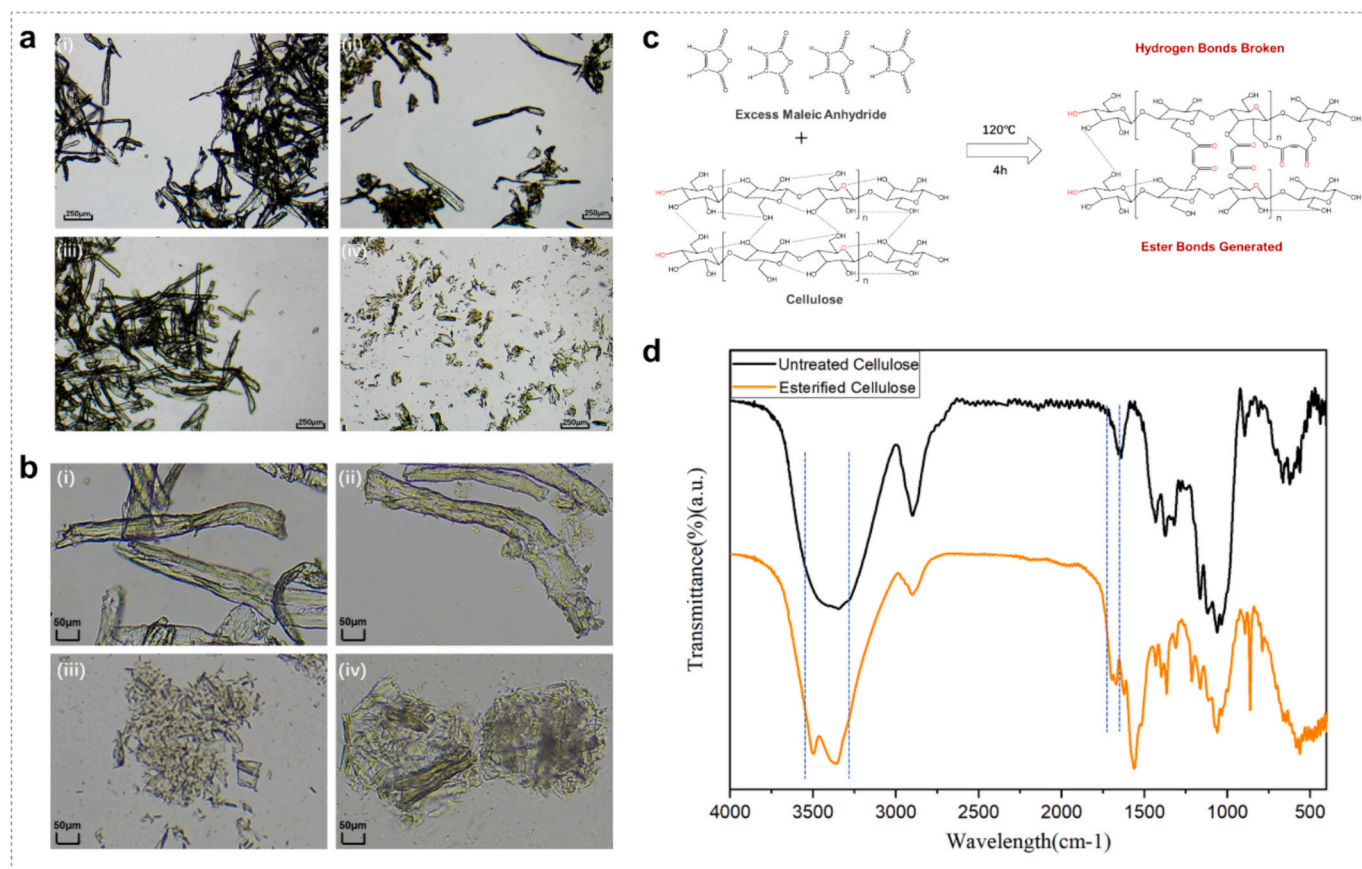


Fig. 1. a) OM of i) untreated, ii) wet maleic anhydride esterified, iii) alkali-treated and iv) semi-wet maleic anhydride esterified. b) Surface structure of i) untreated, ii) alkali-treated, iii) semi-wet maleic anhydride esterified and iv) alkali-ester treated cellulose. c) Schematic diagram of cellulose esterification reaction. d) FT-IR spectra before and after cellulose treatment.

with specific properties are introduced on the surface of cellulose to break the hydrogen bonds between cellulose molecules (Fig. 1(c)), reducing the particle size of cellulose. Due to the larger amount of water added to wet esterification, part of the maleic anhydride was hydrolyzed into maleic acid, then esters and water were produced, and water will hydrolyze the ester group. The semi-wet esterification directly reacted with anhydride, then esters and acids were produced, and the ester groups would not be acidolysis, and the reaction proceeded more completely. Therefore, the dispersion is the most uniform and the particle size is minimal. Decreasing the particle size can potentially increase transparency, since small particles appear less turbid and scatter light more homogeneously [33]. At the same time, esterification it also prevents the formation of clusters and the coating swelling due to the presence of too many hydroxyl groups on the surface of the fiber [32]. The fibers modified by maleic anhydride esterification weakened the elongation vibration peak of the hydroxyl group at about $3600\text{--}3300\text{ cm}^{-1}$; At $1750\text{--}1735\text{ cm}^{-1}$, an ester-based stretching vibration peak occurs (Fig. 1(d)). Maleic anhydride reacts with more hydroxyl groups on cellulose to form cellulose esters and maleic acid, and the acid will not hydrolyze the ester group, and then neutralizes the unreacted anhydride and the acid produced with a low concentration of sodium hydroxide. The hydroxyl reaction on cellulose is reduced, producing cellulose esters, breaking the hydrogen bonds between cellulose molecules, prompting their dispersion. Furthermore, alkali-ester treated cellulose failed to display significant advantages in the cellulose roughness and particle size, compared with the semi-wet esterification treatment (Fig. 1(b) (iv)). To simplify the experimental procedure, semi-wet esterification was finally selected as the cellulose pretreatment scheme.

3.2. Fabrication of saxifrages-biomimetic SES Superhydrophilic coating

Saxifrages are hydrophilic with rough leaves adorned with elongated, delicate tomentum and bumpy cell wall that efficiently capture and spread water droplets. At the same time, wax-free leaf and hydrophilic stomata endow them with high surface energy for rapid water absorption (Fig. 2(a)) [27]. To mimic the high hydrophilicity and rough structure of Saxifrage, cellulose, derived from plants, acted as coating

fillers to mimic tomentum; waterborne styrene-acrylate latex (SAL) simulated the leaf (Fig. 2(a)). However directly adding EC into styrene-acrylate latex (SAL), esterified cellulose (EC) would be encapsulated in the coating after drying (Fig. 2(b) (i-ii)). The EC content exceeded 40 wt %, the EC lacked compatibility and dispersibility of in the SAL system, leading to settling in coatings, due to the gravity. With the addition of sodium dodecyl sulfate (SDS) extracted from coconut oil, EC uniformly dispersed and exposed on the coating surface (Fig. 2(b) (iii-iv)). Since the particle of SAL was decreased in size and increased in number by the SDS, the viscosity of SES decreased, hindering the normal film formation [34]. Affected by polarity, cellulose is embedded in the interstices of these SAL particles, oriented to form a rough surface mimicking the Saxifrages (Fig. 2(c)). This coating was named SES (SAL/EC/SDS) coating. The obtained SES coating exhibited superhydrophilicity with water contact angle of almost 0° and underwater superoleophobic with underwater oil contact angle of about 157.1° , respectively. The results indicated that the Saxifrages-biomimetic coating had excellent surface superhydrophilic properties. (Fig. 2(d)).

3.3. Superhydrophilicity and mechanical properties of saxifrages-biomimetic SES Superhydrophilic coating

Analyzing the static contact angle (WCA) of the coating, both EC and SDS can increase hydrophilicity of coating, the SES coatings especial showed super-hydrophilicity of almost 0° WCA. With the surface free energy of coating increased, the WCA decreased (Fig. 3(a)), elucidating a positive correlation between the WCA of the coatings and their surface free energy., the coating surface tension of water is, while the surface free energy of SES coating is 70.46 dynes/cm , which is close to the maximum tension of water (72 dynes/cm , under ambient temperature [15]). The hydrophilic properties of the coating are mainly determined by roughness and surface energy. Thus, esterified cellulose (EC) was selected as the filler. Although cellulose esters themselves are insoluble in water, the hydroxyl groups in their molecules are still able to form hydrogen bonds with water molecules. The functional groups on the EC are hydrophilic groups [35], and the exposed EC increased the roughness of the surface [29], which are both beneficial for efficient super hydrophilic property. Meanwhile, the surface free energy of the pure

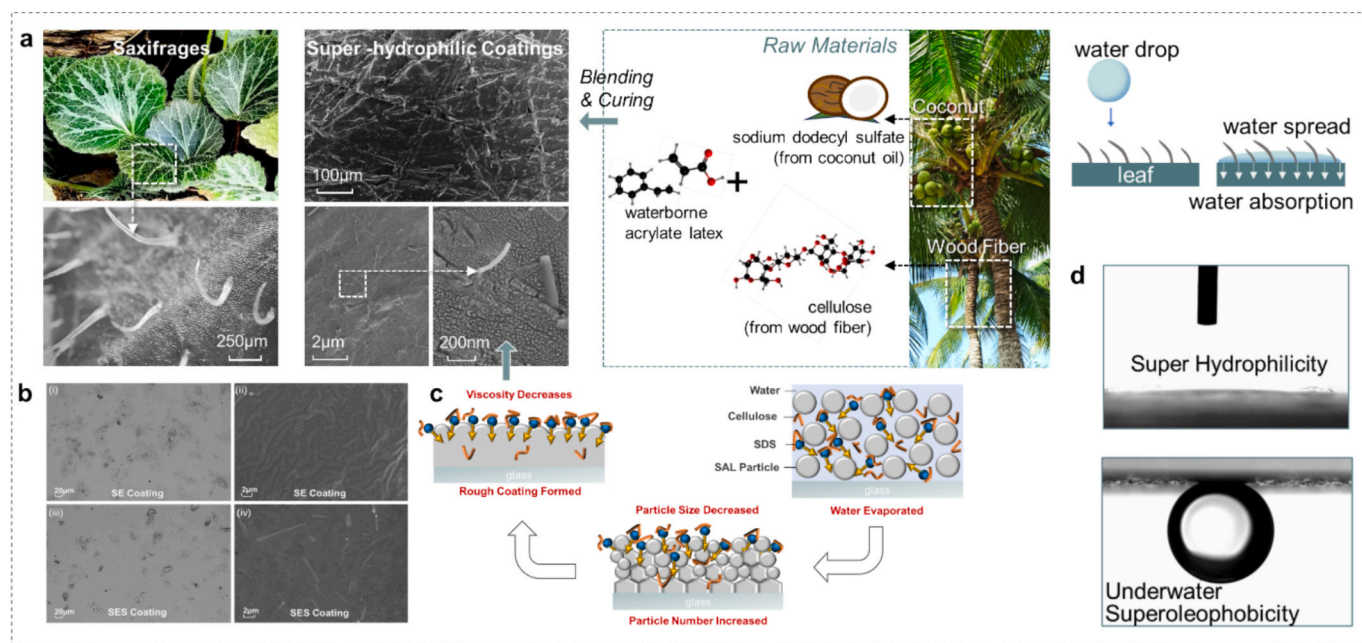


Fig. 2. a) Surface structure of Saxifrages and of SAL/EC/SDS (SES) coating, composition of SES coating, schematic diagram of water droplets falling on the surface of Saxifrage. b) i) OM and ii) SEM images of SAL/EC (SE) coating. iii) OM and iv) SEM images of SES coating. c) Schematic diagram of the film-forming process of SES coating. d) Water contact angle and underwater oil contact angle of the SES coating.

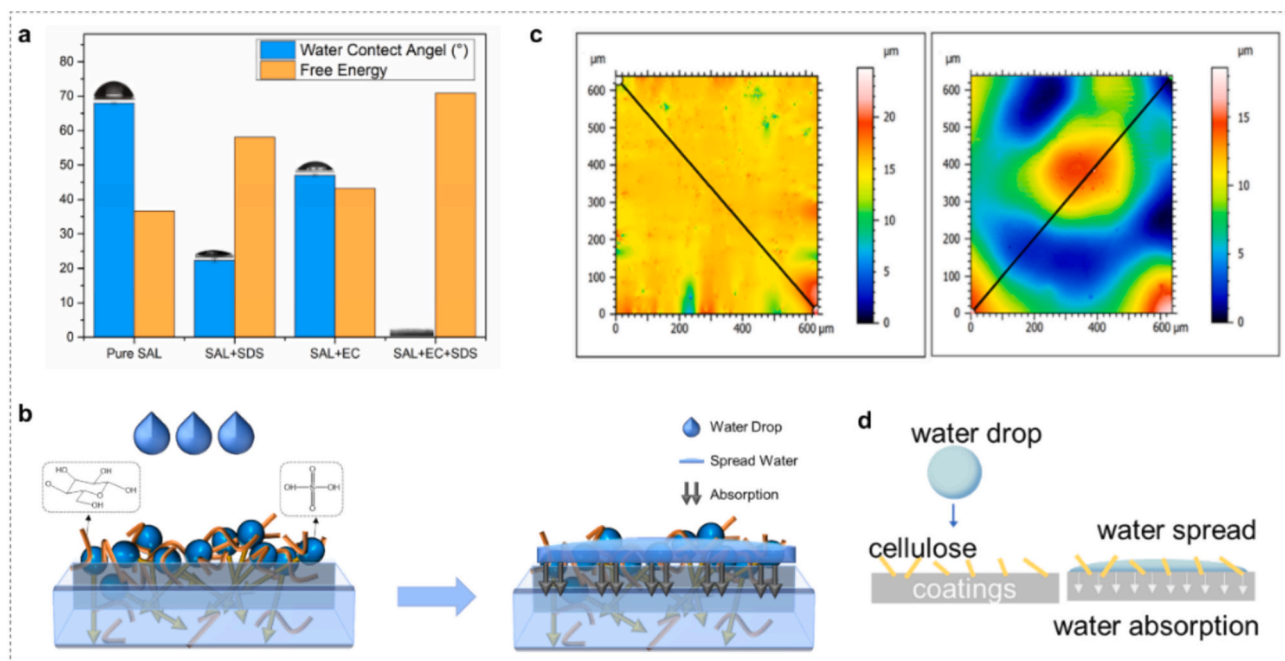


Fig. 3. a) Static water contact angle and free energy of coatings with different components. b) Schematic diagram of hydrophilic mechanism. c) Surface roughness of i) pure styrene acrylic coatings and ii) SES coating. d) Schematic diagram of water droplets falling on the surface of the SES coating.

styrene-acrylate latex (SAL) coating was 36.69 dynes/cm (Fig. 3(a)), which is half that of the SES coating, and the absolute value of zeta potential of the SAL was 0.6, deduced that it was non-polar. Because SDS with hydrophilic and hydrophobic groups reduced surface tension of liquids, dispersed and emulsified liquids. SDS can improve the polarity of coating, due to the molecular structure of SDS was large hydrophilic heads with short hydrophobic tails. Therefore, SDS hydrophobic tail groups inserted into non-polar SAL, and the hydrophilic heads with negative charges repelled each other and tended to face the surface of liquid (Fig. 3(b)). Therefore, after coating was dried out, SDS heads would have been adsorbed onto the coating surface, introducing additional charges [36]. This elevated the charge density on the coating surface, thereby scaling up the electrostatic potential energy to enhance surface energy of the coating surface [37]. The effects of different surfactants on the wettability of coatings were investigated (Fig. S2). It was found that 6 wt% of SDS had the best wettability performance (Fig. S3). It was proved that both EC and SDS can increase the hydrophilicity of composite coatings, because of synergy between the hydrophilic hydroxyl group on the EC and the sulfate group on the SDS head group. Since water, with its relatively lower surface tension, formed a smaller contact angle at the solid-liquid interface, superhydrophilic materials, with their high surface energy, can attract water molecules expeditiously. When dropped on the hydrophilic EC of the SES coating, water penetrated between the EC and SDS and spread continuously, then absorbed by the coating (Fig. 3(d)). This attributed to the presence of hydrophilic hydroxyl groups on the EC surface and hydrophilic head groups on SDS, which provided more water molecule adsorption sites.

Furthermore, laundry detergent is a powdered synthetic detergent containing anionic and nonionic surfactants, which is widely used in industrial or household washing products due to its excellent cleaning effect. When 18 wt% of laundry detergent powder, as a replacement for SDS powder, was added to the coating system, the contact angle of coating can also be basically 0° (Fig. S4), providing a suitable solution for industrial applications.

The root mean square of surface roughness (Rq) of the SES coating is 23.43 μm, which is approximately 10 times greater than the Rq of the pure styrene-acrylate latex (SAL) coating which is 2.58 μm (Fig. 3(c)). In addition to high surface energy, surface roughness is also a crucial factor

in determining the hydrophilicity of the coating [15]. A rough hydrophilic surface facilitates the rapid spreading of water droplets (Fig. 3(d)). Consequently, characterizing the roughness of the coating is particularly important.

According to Wenzel's equation:

$$\cos \theta^* = r \cdot \cos \theta$$

θ^* is the apparent (roughness) contact angle, θ is the Young's contact angle (ideal smooth surface, i.e., $r = 1$), and r is the roughness.

For a hydrophilic surface (with a water contact angle less than 90°), as the roughness increases, the water contact angle decreases, and the surface becomes more hydrophilic.

Therefore, the bio-based SES coatings exhibited superhydrophilic due to the synergistic effect of high surface energy and surface roughness brought about by esterified cellulose (EC) and SDS. The EC and SDS are bonded by waterborne SAL film, it provides a substrate for water absorption, similar to Saxifrage leaves.

Moreover, the increase in hardness is expected to reduce the coating wear rate, potentially improving the mechanical properties of the coating [38]. The tensile strain-stress curves of SES and pure styrene-acrylate latex (SAL) films are shown in Fig. 4(a), displaying that the tensile strength of SAL film was about 0.6 MPa, while the tensile strength of SES coating film was about 1.8 MPa, three times higher than that of the SAL film. The SES coating was able to survive 200 cycles of abrasion under the 50 g load, and remained superhydrophilic after 600 cycles under the same applied load (Fig. 4(b)). There was significant wear until 1350 cycles (Fig. S5). Fig. 4(c) illustrates the excellent adhesion of the SES coating compared to the SAL coating. To a large extent, the strong mechanical strength of SES film is formed by introducing esterified cellulose (EC) as an enhancement to the SAL and the hydrogen bonding between EC and SAL during water removal and the friction force of the entangled network generated during film formation [38].

3.4. Antifogging application of saxifrage-biomimetic SES Superhydrophilic coating

Changes in ambient humidity and temperature making device

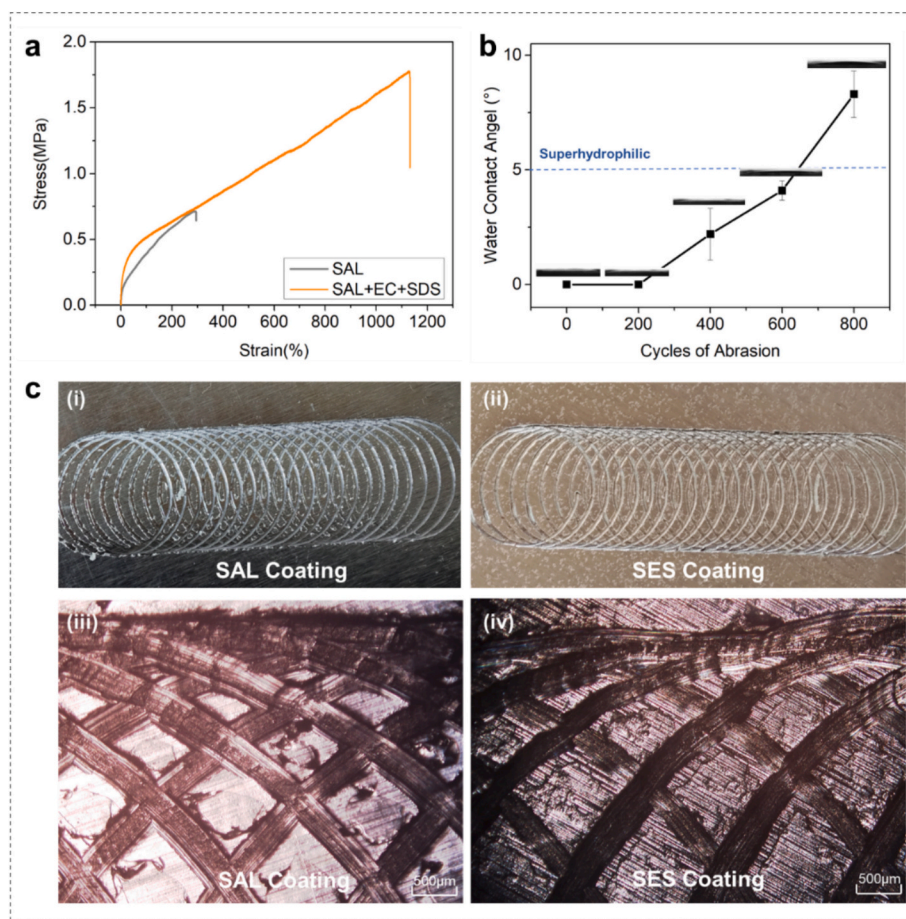


Fig. 4. a) Stress-strain diagram of pure SAL and SES (SAL/ EC/ SDS) coating. b) Abrasion resistance test of SES coating. c) Adhesion testing of pure SAL and SES coating.

fogging is a common phenomenon [39–40]. After the steam test, uncoated glass became blurred and the styrene-acrylate latex (SAL) coated glass fogged severely (Fig. 5(a) (i–ii)), the water contact angle (WCA) of the SAL coating was about 46° (Fig. S1 (a)). Comparatively, the SES coated glass (WCA nearly 0° , shown in Fig. 2(d)) exhibited high transmittance with the words behind clearly visible, indicating remarkable anti-fogging capability of the SES coating, which effectively prevents the accumulation of water condensation on the glass surface (Fig. 5(a) (iii)).

The transparency of devices, such as glasses, car mirrors, goggles and solar sensor, is significantly reduced due to the strong scattering of incident light by condensate droplets because of fogging, bringing great inconvenience to practical applications [41]. For further application, the SES coatings were applied to substrates of different materials such as car mirrors, goggles and eyeglass lenses. The SES coated devices demonstrated excellent anti-fogging performance (Fig. 5(b)–(d)). Given that the transmittance of SES in the visible light range (400–800 nm) is higher than 95 % (Fig. 5 (e)), it can be applied to optics. To prove this, SES anti-fogging coating was applied to the solar panels of solar lamps. It was found that the irradiation intensity of the uncoated panel was 101.3 W/m^2 , and the SES coated panel was 115.2 W/m^2 (Fig. 5(f)). This means that SES anti-fog coating can reduce fogging and dew on the surface of the solar sensor, thereby improving the energy efficiency of the solar panels. It also protects the solar sensors from moisture and other environmental factors, thereby extending the lifespan of the solar panels and reducing maintenance costs [23] [42]. For the SES coating, micro-nano hierarchical surface structure provides a large number of sites for water molecules to adhere to, further enhancing hydrophilicity. This synergy effect of high surface energy and rough surface structure results in the exceptional ability of superhydrophilic materials to absorb water [9],

[18]. It can resist fogging through rapid diffusion and absorption (Fig. 3 (d)). Therefore, antifogging is a typical application for our SES coatings.

3.5. Anti-biofouling ability of saxifrage-biomimetic SES Superhydrophilic coating

Biofouling on the coating surface significantly effects on the service life and cost-effectiveness of the materials. Generally, the formation of biofilms composed of organic compounds, such as proteins and marine algae, is considered to be the initial phase of biofouling [43], [44]. Thus, the anti-biofouling abilities were investigated through static adsorption with various types of biological contaminants, encompassing fluorescein isothiocyanate-labelled bovine serum albumin (FITC-BSA) and *cyclotella cryptica* (CC). The SES coating exhibited significantly lower adsorption levels of FITC-BSA or CC compared to the glass substrate and the SE coated (which consists of styrene acrylic latex and esterified cellulose) one (Fig. 6(a)–(b)). The SES coating showed resistance to contaminants. It is also worth noting that a substantial amount of CC was deposited in certain areas on the SE coating (Fig. 6(b)(ii)).

The minimal adsorption of BSA and CC can be attributed to the synergetic influence of the surface's hydrophilic characteristics and its micro-nanoscale structural features of the SES coating. Hydrophilic groups, such as hydroxyl, can effectively hinder the adhesion of proteins and algae, due to their ability to form a stable and strongly hydrated surface on the surface of the material [15]. Hydrophilic esterified cellulose (EC) with micro-nano structure increases the hydrophilicity and surface roughness of the SES coating, forming a hydration layer to establish primary stain resistance. The small particle size EC ensured that the SES coating had a higher surface roughness, which maintained

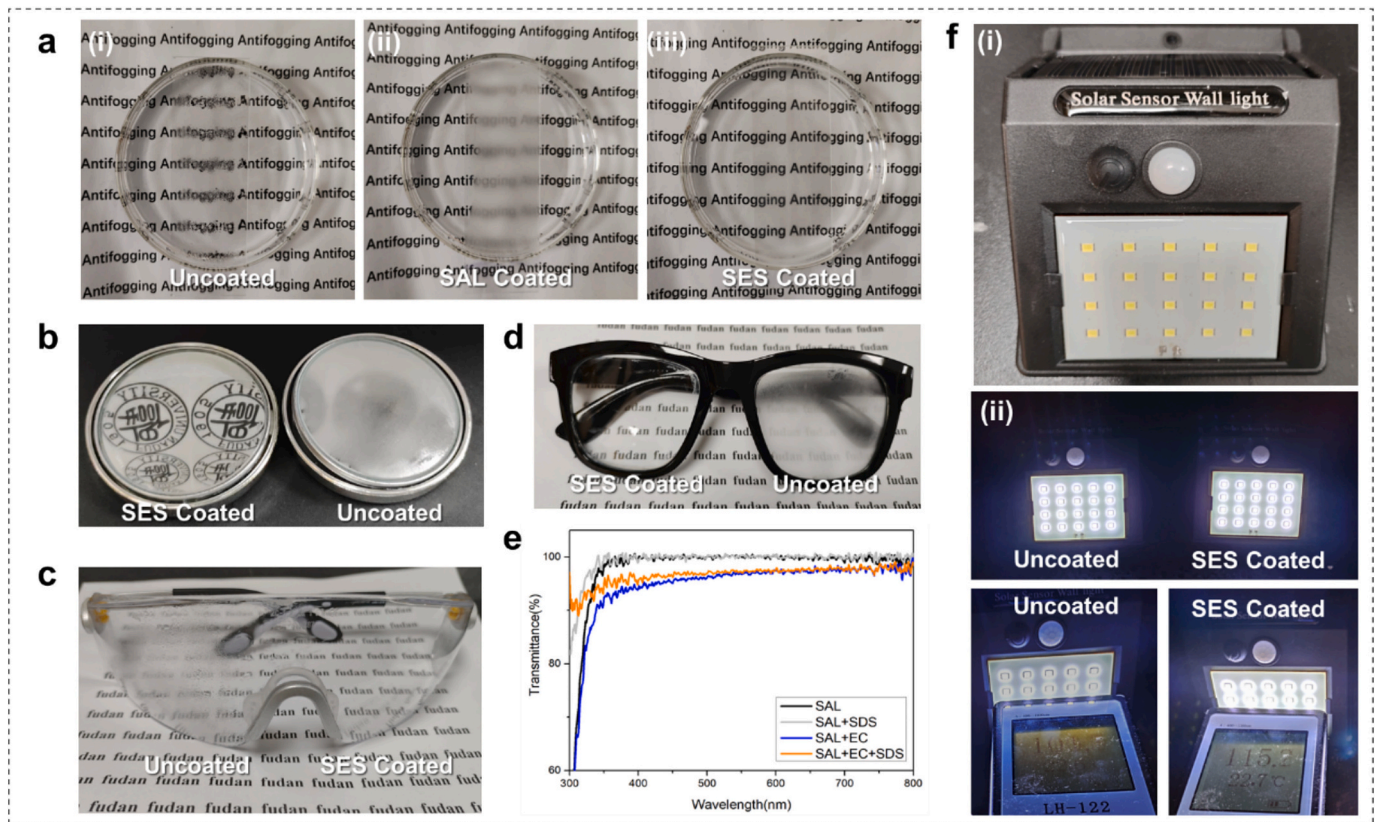


Fig. 5. a) i) uncoated glass, ii) SAL coated and iii) SES coated glass holding over hot water. Uncoated and SES coated b) car mirrors, c) goggles and d) glasses holding over hot water. e) Visible light transmittance of coatings with different components. f) Irradiation intensity of uncoated and SES coated solar sensor wall light after fogging test.

its superhydrophilicity. Meanwhile, the tomentum structure increased the contact area between the water molecules and the coating surface, further generating a dense hydration layer and increasing the steric hindrance interaction between the membrane surface and the oil molecules. The dense hydration layer and steric hindrance complement each other to inhibit the initial adsorption, migration and deposition of biofouling on the surface [29]. However, for the SE coating, EC was encased in the coating without sodium dodecyl sulfate (SDS), resulting in less exposure of the hydrophilic groups on the EC surface. Thus, the hydration layer was unable to form and biofouling gathers in the grooves of the coating (Fig. 6(c)). In addition to hydrophilic surfaces, the other reason for BSA basically non-adsorption was the synergistic effect of SDS. SDS formed a monolayer in an aqueous solution and formed hydrogen bonds with water molecules, making its hydrophilic carboxyl group outward and hydrophobic alkyl group inward [39]. When SDS interacted with proteins, the hydrophilic carboxyl groups of SDS form electrostatic interactions with polar amino acid residues in proteins (such as glutamic acid, aspartic acid, etc.). The hydrophobic alkyl groups of SDS interacted with non-polar amino acid residues (such as alanine, phenylalanine, etc.) in proteins, destroying the tertiary structure of the BSA and linearizing it (Fig. 6(d)). This interaction between SDS and proteins, often referred to as 'hydrophobic interaction', influences the adsorption capacity of a protein, which is related to its structure and properties [45]. When the structure of a protein is disrupted, its adsorption capacity may be affected [46–47]. In sum, the hydration layer caused by hydrophilic group, as well as the synergistic effect of SDS, resulted in effective biofouling resistance of the SES coating. Therefore, the application of SES coating to ships can effectively prevent marine organisms from adsorbing on the bottom of the ship, which causes excess fuel consumption and coating failure, reducing the cost of sailing and maintenance.

3.6. Oil/water separation ability of saxifrage-biomimetic SES Superhydrophilic coating

Advances in offshore oil extraction techniques have inadvertently escalated the risk of marine oil spills, posing significant threats to both aquatic ecosystems and human societies [14]. Innovative methods such as the use of superhydrophilic coatings are needed to solve the problem. A water oil separation test was conducted using the SES mesh, which was prepared by dip-coating the mesh film into SES blended solution (Fig. 7(a)). With the device presented in Fig. 7(b), the ability of SES coated mesh to absorb water and separate oil from an oil–water mixture was investigated. The oil–water mixture was prepared with a 1:4 oil-to-water ratio, simulating the water-to-oil ratio of an oil spill. Following the separation process, the water–oil separation rate was calculated. Fig. 7(c) illustrated the SES coating separation rate, and oil–water separation efficiency (E):

$$E = [m_2 - (m_3 - m_1)] / m_2 \times 100\%$$

m_1 is the initial water quality; m_2 is the initial oil quality; m_3 is the amount of remaining water in the container below.

The SES mesh showed a high water–oil separation capacity, the separation efficiencies for several oils were all over 95 %, especially, the separation efficiency of hexadecane reached 98.5 % with an underwater contact angle for n-hexadecane of 157.1° (Fig. 2(d)), which means underwater superoleophobic.

Compared to other similar coatings. The SES coating exhibited excellent oil/ water separation efficiency (Fig. S6). This is because of the synergistic effect of hydrophilic groups and surface bio-inspired microstructure on the SES. The principle of water–oil separation rests on the superhydrophilic surface's ability to promote the rapid spreading of water while repelling oil substances [22]. Due to polarity, the water

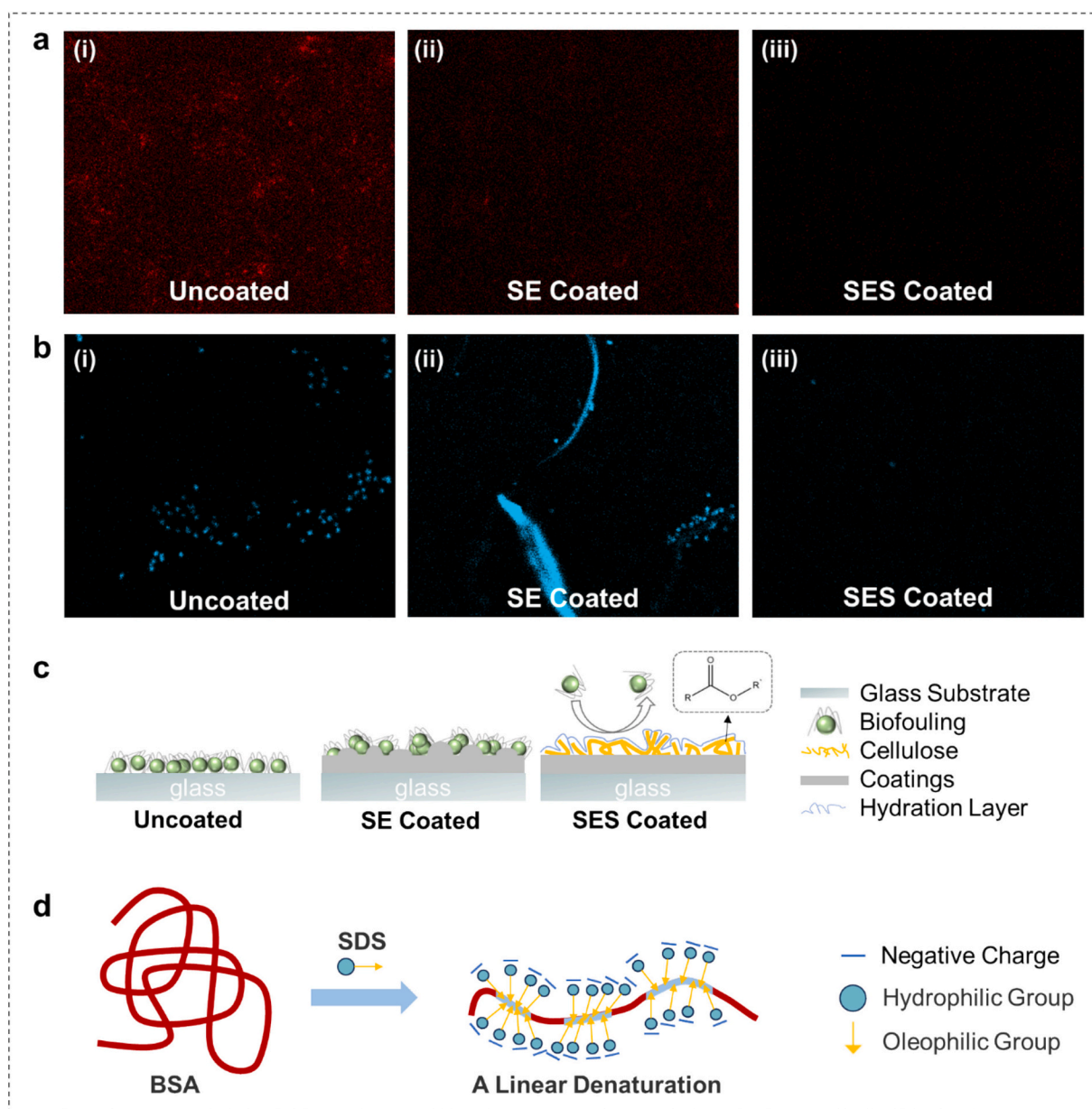


Fig. 6. i) uncoated glass, ii) SAL coated and iii) SES coated glass soaked in a) BSA and b) CC. c) Schematic diagram of the anti-biofouling mechanism of the SES coating. d) Schematic diagram of SDS lysis of BSA.

molecules formed hydrogen bonds with the hydrophilic groups (the hydroxyl group on the esterified cellulose (EC) and the sodium dodecyl sulfate (SDS) sulfated group) on the SES mesh, thus spread out rapidly (Fig. 7(d)). This reduced the mutual attraction between water molecules and makes water easier to pass through the mesh. On the contrary, oil droplets tended to remain drop-like due to their non-polar nature, low surface tension and no strong interactions formed with hydrophilic groups. Besides the chemical composition on SES mesh, the efficient separation is also related to the microscopic morphology structure [48]. For liquids with high surface tension (such as water), oil is liquid with low surface tension and generally exhibits a small solid-liquid interface contact angle. When being dropped, the 'tomentum' structure of SES mesh would remain oil agglomerate above the SES mesh (Fig. 7(d)), achieving water/oil separation. In addition, the SES mesh also has a certain separation effect on the more stable oil-in-water emulsion (Fig. S7), offering an effective solution for industrial applications. The SES mesh is composed of low-cost bio-based materials, with its excellent water/oil separation capabilities, providing a safe, environmentally

friendly and low-cost treatment solution for wastewater treatment and offshore oil spills.

4. Life cycle assessment (LCA) analysis of saxifrage-biomimetic SES Superhydrophilic coating

To obtain the environmental impact of saxifrage-biomimetic superhydrophilic cellulose-based (SES) coating in the full life cycle, the life cycle assessment (LCA) analysis of SES coating, pure styrene-acrylic latex (SAL) coating, and solvent-based superhydrophilic triton X-100/epoxy composite (TXE) coating was investigated. Fig. 8. was the schematic of raw materials, production, and application processes of SES coating. The main raw materials of the SES coating, cellulose and SDS were extracted from trees and coconut oil, respectively (Fig. 8 (a)). The SES coating was prepared by one-step method. Esterified cellulose was added to SAL with SDS, being stirred evenly, coated and film-formation (Fig. 8 (b)). Water was used as solvent. The process was simple and energy-saving, with non-pollution. And the SES coating could be widely

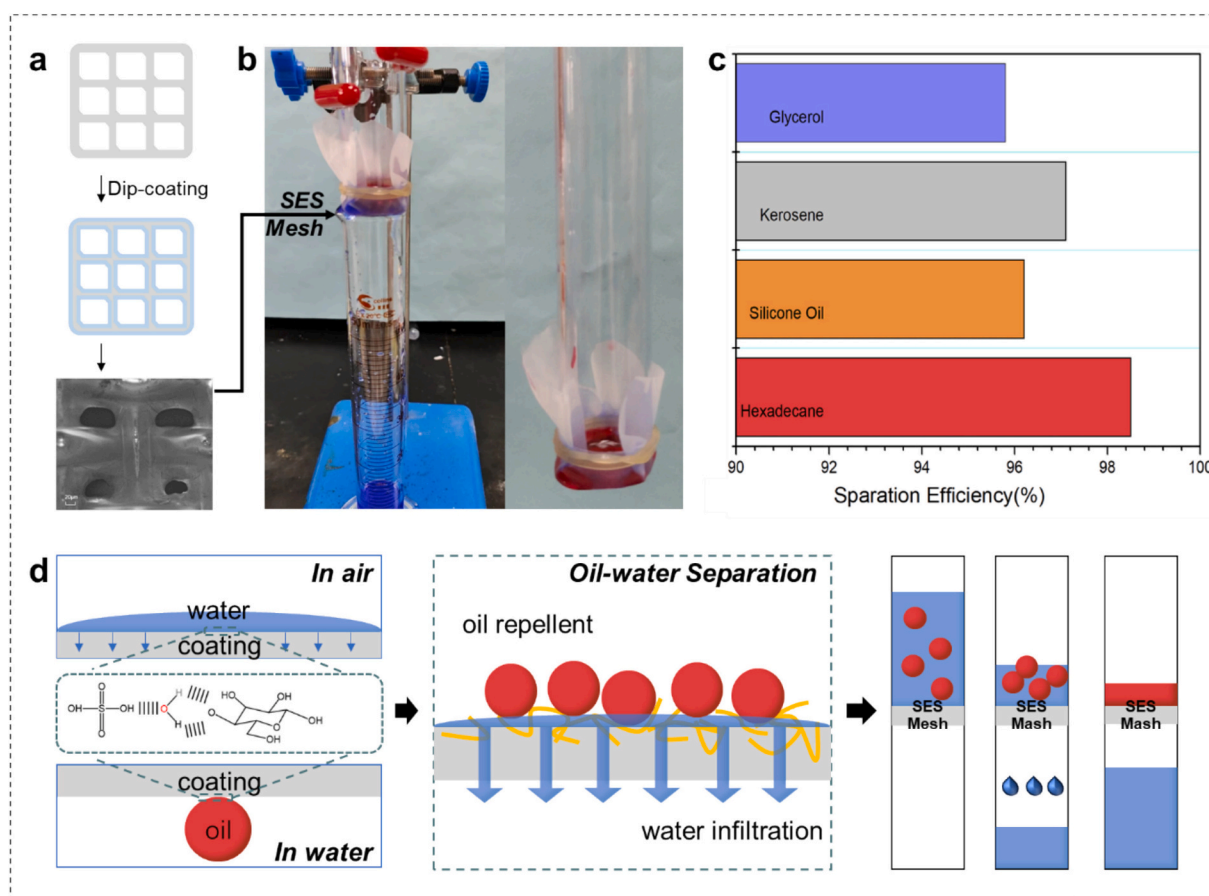


Fig. 7. a) Preparation schematic diagram and SEM images of the SES mesh. b) Water and oil separator. c) Separation efficiency of the SES coatings for different oils. d) Schematic diagram of the water-oil separation mechanism of the SES coating.

applied. The SES coating can be applied to glasses, automotive glass, and rearview mirrors, which can prevent obstruction of vision caused by fogging and reduce the rate of traffic accidents. When applied to solar panels, it can prevent the condensate droplets formed by mist from scattering the incident light and improve the solar energy conversion rate. The adsorption of marine organisms such as algae will increase the weight of the ship, thereby increasing fuel consumption, and cleaning up the marine organisms adsorbed on the ship is also time-consuming and costly. The SES coating provides a solution for anti-biofouling on ships and reduces sailing costs. The ocean is rich in petroleum energy, but if it leaks during extraction or transportation, it will seriously affect the marine ecological environment. Thus, the SES coating with 98.5 % water-oil separation efficiency and low cost of are suitable for dealing with this problem. (Fig. 8 (c)).

Table 1 showed nine key points of environmental impact categories on coatings, including bio-based content of coating, production process, solvent content, VOC content, toxic and harmful substance content, retrofitting of equipment and painting lines, energy consumption in painting, multifunctional application, degree of degradability, respectively. The SES coating with about 51 % bio-based content was partly degradable. While the TXE coating was fabricated by organic solvent and diglyceryl ether of bisphenol A. Therefore, the amount of volatile organic compounds (VOC) and harmful substance was much higher than that of SES coating. None of these three coatings need to retrofit the equipment and painting lines. As painted directly, they all consumed a little energy during painting. The SES coating had a wider range of applications than these two coatings, especially compared to SAL coating.

According to environmental impact categories on coatings, life cycle assessment (LCA) analysis was listed as a score form, shown in Table 2.

Compared to the TXE and SAL coating, our SES coating exhibited excellent environmental performance (90 points) in the life cycle assessment (LCA) analysis. Because it was made of low-cost bio-based materials (cost of raw material seen in S9), energy saving and non-pollution during production and painting, as well as wide applications. Therefore, it can be concluded that the SES coating reduced the environmental burden throughout the whole life cycle, and would have a promising future in driving cleaner production and sustainability goals.

5. Conclusions

In this work, a cellulose-based superhydrophilic coating (SES) was fabricated via mimicking the Saxifrage leaf structure through blending semi-wet maleic anhydride esterified cellulose (EC) with waterborne styrene-acrylate latex (SAL) and sodium dodecyl sulfate (SDS). Compared to several modification methods of cellulose, the semi-wet esterified cellulose (EC) was chosen as filler of coating because of its smallest particle size and the best dispersion of cellulose in waterborne latex with the least energy consumption.

Due to the addition of hydrophilic sodium dodecyl sulfate surfactant (SDS), needlelike cellulose was orientally arranged in coating to imitate saxifrage structure. The superhydrophilicity of coating was obtained by the synergy effects of the micro-nano scale roughness, high surface energy and hydrophilic group. The water contact angle of the SES coating was almost 0° and the underwater contact angle was about 157.1° . The cellulose endowed the SES coating with a tensile strength of 1.8 MPa, three times that of pure styrene-acrylate latex (SAL) coating and provided ability to resist abrasion. The SES coating remained superhydrophilic after 600 cycles of abrasion.

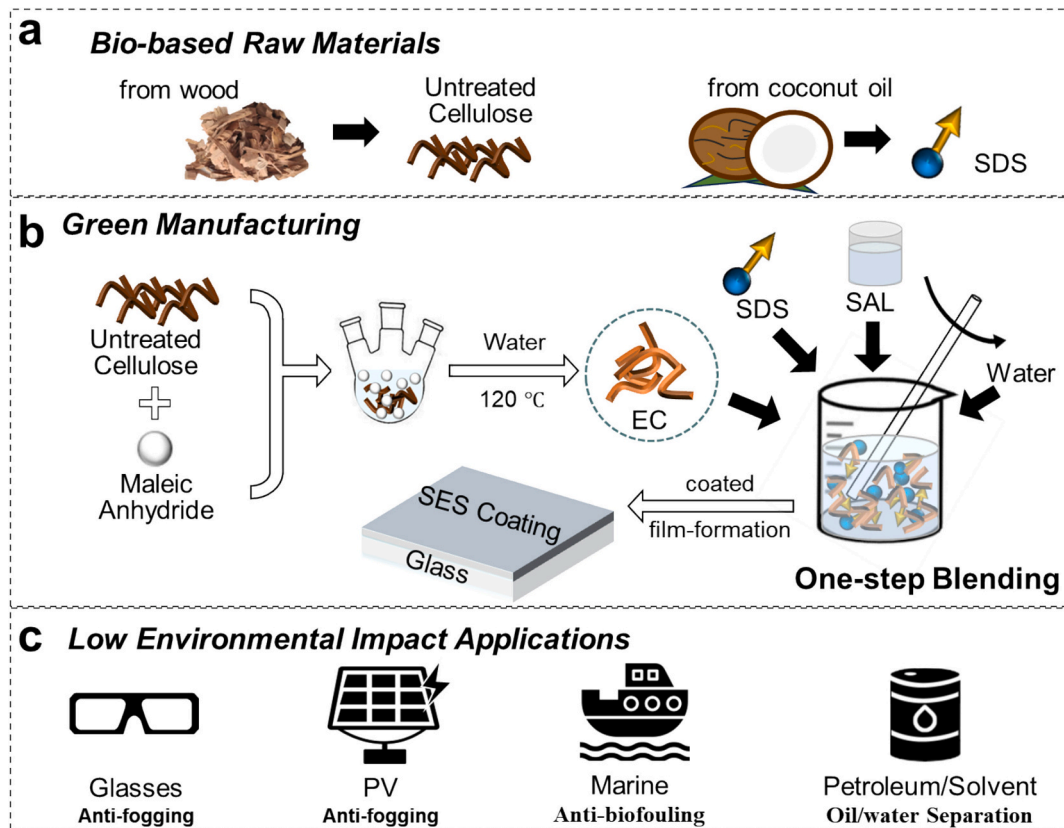


Fig. 8. The full life cycle of SES coatings.

Table 1
Environmental impact categories on coatings.

Environmental Impact Categories	TXE Coating [2]	SAL Coating	SES Coating (This work)
Bio-based Content of Coating	0 %	0 %	51 %
Production Process	One-step preparation	Direct preparation	One-step preparation
Solvent	Organic solvent	Water	Water
VOC Content and Emissions	50 g/L	0 g/L	0 g/L
Toxic and Harmful Substance Content	Over 50 %	0 %	0 %
Retrofitting of Equipment and Painting Lines	No need	No need	No need
Energy Consumption in Painting	Painting directly	Painting directly	Painting directly
Multifunctional Application	Anti-fogging Oil/water separation	No application	Anti-fogging Anti-biofouling Oil/water separation
Degree of Degradability	0 %	0 %	Over 50 %

Table 2
Score of life cycle assessment (LCA).

Environmental Impact Categories*	TXE Coating [2]	SAL Coating	SES Coating (This work)
Raw Materials (x2)	10	10	16
Energy Consumption in Production Process	8	10	8
Cleaner Production	5	8	8
VOC Content and Emissions	5	10	10
Toxic and Harmful Substance Content	5	10	10
Retrofitting of Equipment and Painting Lines	10	10	10
Energy Consumption in Painting	10	10	10
Multifunctional Application (Anti-fogging, Anti-biofouling, Oil/water Separation)	8	1	10
Degree of degradability	1	1	8
Score	62	70	90

* Note: Score for 4 grades: 10 points are the best, 8 points are good, 5 points are neutral, and 1 point is the worst. The raw material is an important indicator of bio-based coatings, the score x2.

The SES coating proved to be outstanding in anti-fogging, anti-biofouling and oil/water separation properties. The SES coating with 95 % transmittance and anti-fogging property can be covered on solar panels to increase the energy conversion rate. The SES coating also showed high biofouling resistance against *cyclotella cryptica* (CC) and bovine serum albumin (BSA), attributed to the hydration layer, formed by hydrophilic groups with water. The presence of SDS was also key to cleave BSA. Thus, it can be applied to reduce fuel consumption and cleaning costs for ocean vessels caused by the adsorption of marine organisms. Furthermore, dip-coating SES mesh presented over 95 % water-oil separation efficiency of glycerol, kerosene, silicone oil and

hexadecane. Due to the polar water molecules formed hydrogen bonds with the hydrophilic groups on the SES coating, spreading out rapidly. While the non-polar oil molecules were further blocked by the micro-nano structure formed by the EC. It provides a solution to reduce the pollution of the ocean caused by oil leakage in the extraction and transportation of oil in the ocean. As low cost and green cellulose-based, the SES superhydrophilic coating reduced carbon emission in full life cycle. For wide applications, it can balance economy, environmental friendliness and practicality.

CRediT authorship contribution statement

Yan Xu: Writing – original draft. **Qingyu Wu:** Writing – review & editing. **Dicong Yang:** Writing – review & editing. **Biwei Qiu:** Writing – review & editing. **Xiaoliang Mo:** Writing – review & editing. **Bo You:** Formal analysis, Conceptualization. **Limin Wu:** Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.susmat.2025.e01278>.

Data availability

All data needed to evaluate the conclusions in the paper are presented in the paper and/or the Supplementary Information. Additional data related to this paper may be requested from the authors.

References

- X. Guo, S. Ge, J. Wang, X. Zhang, T. Zhang, J. Lin, C.X. Zhao, B. Wang, G. Zhu, Z. Guo, Waterborne acrylic resin modified with Glycidyl methacrylate (GMA): formula optimization and property analysis, *Polymer* 143 (2018) 155–163.
- A. Caretto, V. Passoni, N. Brenna, M. Sitta, L. Oglioni, G. Catel, S. Turri, G. Griffini, Fully biobased polyesters based on an isosorbide monomer for coil coating applications, *ACS Sustain. Chem. Eng.* 6 (11) (2018) 14125–14134.
- X. Yue, H.B. Yang, Z.M. Han, Y.X. Lu, C.H. Yin, X. Zhao, Z.X. Liu, Q.F. Guan, S. H. Yu, Tough and moldable sustainable cellulose-based structural materials via multiscale Interface engineering, *Adv. Mater.* 36 (7) (2023) 1–24.
- J.M. Morrisette, P.J. Carroll, I.S. Bayer, J. Qin, D. Waldrup, C.M. Megaridis, A methodology to produce eco-friendly Superhydrophobic coatings produced from all-water-processed plant-based filler materials, *Green Chem.* 20 (22) (2018) 5169–5178.
- S.Y. Soom Goo, S.Y. Park, W. Im, H.J. Youn, Mechanical properties and soiling resistance of paper with polyurethane coating reinforced with cellulose nanomaterials, *Bioresources* 14 (4) (2019) 8973–8986.
- K. Jin, Y. Tang, J. Liu, J. Wang, C. Ye, Nanofibrillated cellulose as coating agent for food packaging paper, *Int. J. Biol. Macromol.* 168 (2021) 331–338.
- P.R. Rajimol, S.B. Ulaeto, A. Puthiyamadam, S.K. Sahoo, T.P.D. Rajan, K. V. Radhakrishnan, R.K. Sukumaran, Development of an oxyresveratrol incorporated bio-based smart nanocomposite coating with anti-corrosive, self-healing, and anti-microbial properties, *Green Chem.* 25 (18) (2023) 7189–7215.
- M.F. Cunningham, J.D. Campbell, Z. Fu, J. Bohling, J.G. Leroux, W. Mabey, T. Robert, Future green chemistry and sustainability needs in polymeric coatings, *Green Chem.* 21 (18) (2019) 4919–4926.
- L. Wen, Y. Tian, L. Jiang, Bioinspired super-wettability from fundamental research to practical applications, *Angew. Chem. Int. Ed.* 54 (11) (2015) 3387–3399.
- S.G.K. Mahdi, M. Muttar, Mohammed E. Ismael, Rashid H. Jabbar, Preparation of TIO₂ PMMA PVA nanocomposite, *J. Phys. Conf. Ser.* 2322 (2022) 012072–012078.
- S. Turkoglu, J. Zhang, H. Dodiuk, S. Kenig, J.A. Ratto, J. Mead, Wetting characteristics of Nanosilica-poly (acrylic acid) transparent anti-fog coatings, *Polymers* 14 (21) (2022) 4663–4677.
- Z. Zeng, Y. Liu, L. Long, J. He, Y. He, L. Wang, G. Yang, X. Zhang, F. Shen, Y. Zhang, Fabrication of novel Superhydrophilic/ underwater Superoleophobic composite coatings and study on the relationship between their Long-term wettability and excellent oil-water separation performance, *Surf. Coat. Technol.* 434 (2022) 128193–128203.
- T. Yuan, J. Yin, Y. Liu, W. Tu, Z. Yang, Micro/nanoscale structured Superhydrophilic and underwater Superoleophobic hybrid-coated mesh for high-efficiency oil/water separation, *Polymers* 12 (6) (2020) 1806421–1806429.
- J.C. Zheng Wan, Human Errors are Behind Most Oil-Tanker Spills, *Nature* 560 (2018) 161–163.
- J. Yong, Q. Yang, X. Hou, F. Chen, Emerging separation applications of surface Superwettability, *Nanomaterials* 12 (4) (2022) 688–746.
- M.Q. Hovish, F. Hilt, N. Rolston, Q. Xiao, R.H. Dauskardt, Open air plasma deposition of Superhydrophilic Titania coatings, *Adv. Funct. Mater.* 29 (19) (2019) 1378–1392.
- J. Li, L. Yang, H. Liu, G. Li, R. Li, Y. Cao, H. Zeng, Simple preparation method for hydrophilic/Oleophobic coatings, *ACS Appl. Mater. Interfaces* 12 (40) (2020) 45266–45273.
- Z. Chu, Y. Feng, S. Seeger, Oil/water separation with selective Superantiwetting/ Superwetting surface materials, *Angew. Chem. Int. Ed.* 54 (8) (2015) 2328–2338.
- J.-B. Chemin, S. Bulou, K. Baba, C. Fontaine, T. Sindzingre, N.D. Boscher, P. Choquet, Transparent anti-fogging and self-cleaning TiO₂/SiO₂ thin films on polymer substrates using atmospheric plasma, *Sci. Rep.* 8 (1) (2018) 9603–9610.
- W. Deng, Y. Su, C. Zhang, W. Wang, L. Xu, P. Liu, J. Wang, X. Yu, Y. Zhang, Transparent Superhydrophilic composite coating with anti-fogging and self-cleaning properties, *J. Colloid Interface Sci.* 642 (2023) 255–263.
- K.A. Henn, S. Babaeipour, S. Forsell, P. Nousiainen, K. Meinander, P. Oinas, M. Österberg, Transparent lignin nanoparticles for Superhydrophilic antifogging coatings and photonic films, *Chem. Eng. J.* 475 (2023) 145965–145980.
- S.N.W. Ikhsan, N. Yusof, F. Aziz, A.F. Ismail, N. Shamsuddin, J. Jaafar, W.N. W. Salleh, W.J. Lau, N. Misdan, Synthesis and optimization of superhydrophilic-superoleophobic chitosan-silica/HNT nanocomposite coating for oil-water separation using response surface methodology, *Nanomaterials* 12 (20) (2022) 3673–3695.
- Intelligent solar-driven “switch” photothermal hydrogel for clean water harvesting, *Chem. Eng. J.* 495 (2024) 15342–15350.
- H. Lai, Z. Ge, X. Li, R. Tao, P. Zhang, Z. Dong, Y. Li, Effect of microstructure and wax on the hydrophobic properties of plant leaves, *Sci. Silvae Sin.* 53 (4) (2017) 74–82.
- C. Yu, C. Li, C. Gao, Z. Dong, L. Wu, L. Jiang, Time-dependent liquid transport on a biomimetic topological surface, *ACS Nano* 12 (6) (2018) 5149–5157.
- W. Pei, J. Li, Z. Guo, Y. Liu, C. Gao, L. Zhong, S. Wang, Yongping Hou, Y. Zheng, Excellent fog harvesting performance of liquid-infused nano-textured 3D frame, *Chem. Eng. J.* 409 (2021) 128180–128187.
- B.A. Bennett, S. Withers, Spiked saxifrage, *Saxifraga Spicata*, rediscovered in Canada after 110 years, *Can. Field-Natural.* 124 (2010) 57–58.
- S. Das, M.J. Hossain, S.-F. Leung, A. Lenox, Y. Jung, K. Davis, J.-H. He, T. Roy, A leaf-inspired photon management scheme using optically tuned bilayer nanoparticles for ultra-thin and highly efficient photovoltaic devices, *Nano Energy* 58 (2019) 47–56.
- Y. Zhao, X. Yang, Z. Cheng, C.H. Lau, J. Ma, L. Shao, Surface manipulation for prevention of migratory viscous crude oil fouling in Superhydrophilic membranes, *Nat. Commun.* 14 (1) (2023) 2679–2690.
- W. Pei, J. Wang, X. Li, Y. Li, K. Ning, J. Wang, Q. Zhang, Multi-dimensional bionic Micro-nanostructure Surface for controllable fog harvesting: from Fabrication to application, *Adv. Sustainable, System* 8 (2024) 2300657–2300670.
- G. Li, J. Fei, Y. Xu, Y. Li, J. Li, Bioinspired assembly of hierarchical light-harvesting architectures for improved photophosphorylation, *Adv. Funct. Mater.* 28 (13) (2018) 1706557–1706563.
- Z.-H. Liu, N. Hao, S. Shinde, Y. Pu, X. Kang, A.J. Ragauskas, J.S. Yuan, Defining lignin nanoparticle properties through tailored lignin reactivity by sequential Organosolv fragmentation approach (SOFA), *Green Chem.* 21 (2) (2019) 245–260.
- X. Wang, Y. Cao, Characterizations of absorption, scattering, and transmission of typical nanoparticles and their suspensions, *J. Ind. Eng. Chem.* 82 (2020) 324–332.
- M.A.E. Hafizah, A.F. Riyadi, A. Manaf, Particle size reduction of polyaniline assisted by Anionic emulsifier of sodium dodecyl Sulphate (SDS) through emulsion polymerization, *PDF, IOP Conf. Series: Mater. Sci. Eng.* 515 (2019) 012080–012087.
- S. Iwamoto, T. Endo, 3 nm thick lignocellulose nanofibers obtained from esterified wood with maleic anhydride, *ACS Macro Lett.* 4 (1) (2014) 80–83.
- N.G.P. Chew, S. Zhao, C.H. Loh, N. Permogorov, R. Wang, Surfactant effects on water recovery from produced water via direct-contact membrane distillation, *J. Membr. Sci.* 528 (2017) 126–134.
- X. Liao, Y. Wang, Y. Liao, X. You, L. Yao, A.G. Razaqpur, Effects of different surfactant properties on anti-wetting behaviours of an omniphobic membrane in membrane distillation, *J. Membr. Sci.* 634 (2021) 119433–119443.
- B. Macdonald, F.-W. Wang, B. Tobelmann, J. Wang, J. Landini, N. Gunaratne, J. Kovac, T. Miller, R. Mosurkal, A. Tuteja, Design of Abrasion-Resistant, Long-lasting Antifog coatings, *ACS Appl. Mater. Interfaces* 16 (10) (2024) 13018–13028.
- H. Yang, L. Bai, Y. Duan, H. Xie, X. Wang, R. Zhang, X. Ji, C. Si, Upcycling corn straw into Nanocelluloses via enzyme-assisted homogenization: application as building blocks for high-performance films, *J. Clean. Prod.* 390 (2023) 136215–136224.
- J. Shi, L. Xu, D. Qiu, Effective antifogging coating from hydrophilic/hydrophobic polymer Heteronetwork, *Adv. Sci.* 9 (14) (2022) 2200072–2200078.
- J. Ren, R. Kong, Y. Gao, L. Zhang, J. Zhu, Bioinspired adhesive coatings from Polyethylenimine and tannic acid complexes exhibiting antifogging, self-cleaning, and antibacterial capabilities, *J. Colloid Interface Sci.* 602 (2021) 406–414.
- C.S. Thompson, R.A. Fleming, M. Zou, Transparent self-cleaning and antifogging silica nanoparticle films, *Sol. Energy Mater. Sol. Cells* 115 (2013) 108–113.
- S.K. Shahrouz Amini, Luigi Petrone, Onyemaechi Ahanotu, Steffi Sunny, Clarinda N. Sultanto, Shawn Hoon, Lucas Cohen, James C. Weaver, Joanna Aizenberg, Nicolas Vogel, Ali Miserez, Preventing mussel adhesion using lubricant-infused materials, *Science* 357 (2017) 668–673.
- J.A. Callow, M.E. Callow, Trends in the development of environmentally friendly fouling-resistant marine coatings, *Nat. Commun.* 2 (1) (2011) 1247–1252.

- [45] D. Winogradoff, S. John, A. Aksimentiev, Protein unfolding by SDS: the microscopic mechanisms and the properties of the SDS-protein assembly, *Nanoscale* 12 (9) (2020) 5422–5434.
- [46] X.G.L. Nicholas, J. Turro, K.P. Ananthapadmanabhan, M. Aronson, Spectroscopic probe analysis of protein-surfactant interactions: the BSA/SDS system, *Langmuir* 11 (1995) 2525–2533.
- [47] M.M. Nielsen, K.K. Andersen, P. Westh, D.E. Otzen, Unfolding of β -sheet proteins in SDS, *Biophys. J.* 92 (10) (2007) 3674–3685.
- [48] C. Chen, D. Weng, A. Mahmood, S. Chen, J. Wang, Separation mechanism and construction of surfaces with special wettability for oil/water separation, *ACS Appl. Mater. Interfaces* 11 (11) (2019) 11006–11027.